

Ex.no:3	DATA CLEANING

Aim:

To clean and preprocess the E-Commerce dataset by identifying and handling missing values, removing duplicate records, detecting outliers, scaling numerical data, encoding categorical data, and preparing a clean dataset for further analysis.

Algorithm:

Check Missing Values

1. Start.
2. Import the required libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.
3. Upload and load the E-Commerce dataset into a DataFrame.
4. Display the first few records, dataset shape, and information.
5. Check for missing values in each column.
6. Fill missing numerical values using the median of the respective columns.
7. Identify and remove duplicate records.
8. Detect outliers in the unit_price column using the IQR method.
9. Visualize the outliers using a boxplot.
10. Apply Min-Max scaling to the unit_price column.
11. Apply Label Encoding to categorical columns.
12. Apply One-Hot Encoding to the category column.
13. Save the cleaned dataset as a CSV file.
14. Stop.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler, LabelEncoder
from google.colab import files
```

```
uploaded = files.upload()

file_name = list(uploaded.keys())[0]
df = pd.read_csv(file_name)

print(df.head())

print("\nDataset Shape")
print(df.shape)

print("\nDataset Information")
df.info()

print("\nMissing Values")
print(df.isnull().sum())

numeric_cols = df.select_dtypes(include=np.number).columns

df[numeric_cols] = df[numeric_cols].fillna(
    df[numeric_cols].median()
)

print("\nMissing Values After Filling")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())

df.drop_duplicates(inplace=True)

print("After Removal:", df.shape)
```

```
Q1 = df["unit_price"].quantile(0.25)
```

```
Q3 = df["unit_price"].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR
```

```
upper = Q3 + 1.5 * IQR
```

```
outliers = df[  
    (df["unit_price"] < lower) |  
    (df["unit_price"] > upper)  
]
```

```
print("\nOutliers")
```

```
print(outliers.head())
```

```
print("Number of outliers:", len(outliers))
```

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(x=df["unit_price"])
```

```
plt.title("Unit Price Outliers")
```

```
plt.xlabel("Unit Price")
```

```
plt.show()
```

```
scaler = MinMaxScaler()
```

```
df[["unit_price"]] = scaler.fit_transform(  
    df[["unit_price"]]  
)
```

```
print("\nScaled Unit Price")  
print(df[["unit_price"]].head())
```

```
encoder = LabelEncoder()
```

```
categorical_columns = [  
    "country",  
    "preferred_category",  
    "product_name",  
    "category",  
    "gender"  
]
```

```
for col in categorical_columns:  
    df[col] = encoder.fit_transform(  
        df[col].astype(str)  
    )
```

```
print("\nEncoded Categorical Columns")  
print(df[categorical_columns].head())
```

```
category_dummies = pd.get_dummies(  
    df["category"],  
    prefix="Category"  
)
```

```

df = pd.concat(
    [df, category_dummies],
    axis=1
)

print("\nOne-Hot Encoded Category")
print(category_dummies.head())

df.to_csv(
    "/content/E_Commerce_Customer_Cleaned.csv",
    index=False
)

print("\nCleaned dataset saved successfully.")

```

Output:

Original DataFrame:

```

Saving E Commerce.csv to E Commerce (2).csv
  order_id customer_id  age product_id  country signup_date \
0  ORD5000  CUST1000   39  PROD200    Canada  01-07-2021
1  ORD5001  CUST1001   61  PROD201     USA  10/19/2020
2  ORD5002  CUST1002   26  PROD202  Pakistan  06-10-2023
3  ORD5003  CUST1003   54  PROD203     India  7/30/2023
4  ORD5004  CUST1004   50  PROD204     India  12-09-2020

  last_purchase_date  cancellations_count  subscription_status  order_date \
0      2/21/2023                0      active  8/20/2024
1      12-08-2021                0      active  7/17/2025
2      09-04-2023                3  cancelled  03-12-2025
3      2/20/2024                4      paused  9/19/2024
4      9/14/2024                0      active  08-08-2024

  unit_price  quantity  purchase_frequency  preferred_category  product_name \
0      78.21         5                37          Sports    Football
1      64.02         8                35          Sports  Refrigerator
2     604.14         2                44  Electronics    Hoodie
3      84.66         4                 1          Sports  Conditioner
4      62.66         7                35          Sports   Smartwatch

  category  gender
0   Sports  Female
1    Home  Female
2  Clothing  Female
3   Beauty   Male
4  Electronics  Male

```

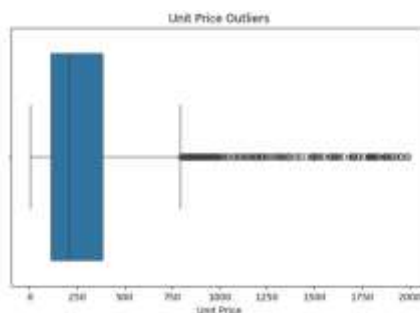
Dataset Shape
(2000, 17)

Outliers

	order_id	customer_id	age	product_id	country	signup_date	\
8	ORD5008	CUST1008	53	PROD208	Pakistan	4/25/2021	
11	ORD5011	CUST1011	24	PROD211	Canada	02-12-2022	
16	ORD5016	CUST1016	34	PROD216	USA	07-06-2021	
18	ORD5018	CUST1018	42	PROD218	USA	5/25/2023	
32	ORD5032	CUST1032	52	PROD232	Germany	02-09-2021	

	last_purchase_date	cancellations_count	subscription_status	order_date	\
8	12/26/2024	4	paused	12/14/2023	
11	03-12-2025	1	active	5/30/2025	
16	08-02-2025	1	active	6/19/2024	
18	01-09-2025	5	paused	8/13/2025	
32	7/27/2021	0	cancelled	11-10-2023	

	unit_price	quantity	purchase_frequency	preferred_category	\
8	1525.80	1	32	Electronics	
11	940.17	1	24	Beauty	
16	1875.08	1	31	Sports	
18	1702.39	1	14	Clothing	
32	804.77	2	26	Beauty	



Scaled Unit Price

	unit_price
0	0.037893
1	0.030758
2	0.302341
3	0.041136
4	0.030074

Encoded Categorical Columns

	country	preferred_category	product_name	category	gender
0	0	4	12	4	0
1	5	4	26	3	0
2	3	2	16	1	0
3	2	4	5	0	1
4	2	4	32	2	1

One-Hot Encoded Category

	Category_0	Category_1	Category_2	Category_3	Category_4
0	False	False	False	False	True
1	False	False	False	True	False
2	False	True	False	False	False
3	True	False	False	False	False
4	False	False	True	False	False

Cleaned dataset saved successfully.

Result: